3. Principles of Ionizing Radiation

This chapter is dedicated to the basic properties of ionizing radiation. The discussion will include a review of atomic structure, radioactivity, the types of radiation, the biological effects of ionizing radiation, and the established maximum permissible doses for the various types of ionizing radiation.

Atomic Structure

Over one hundred chemical elements exist on the earth. These elements provide the basic ingredients in all material things. An atom is the smallest particle of an element that possesses the chemical properties of that element. Each atom is composed of three fundamental particles that are most easily described in terms of two physical properties--electric charge and mass. The first particle is a proton. The proton has a positive charge (+1) and a mass of approximately one atomic mass unit (amu). The identity of the chemical element is based on the number of protons. The second particle is a neutron. This particle has no charge and a mass of approximately one amu. The proton and neutron contribute most of the mass of an atom and reside in a small volume called the "nucleus." The third particle is an electron. The electron has a negative charge (-1) and a mass of approximately 5.4×10^{-4} amu. The electrons exist in orbit around the nucleus. Each orbit represents a higher energy level for the electron.

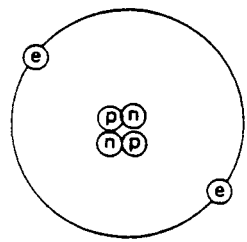
An atom is said to be electrically neutral if the number of protons in the nucleus is equal to the number of electrons orbiting the nucleus. An atom with a surplus or deficit of orbital electrons will be negatively or positively charged respectively and is called an ion. The number of neutrons found in the nucleus is related to the number of protons, but there is not a direct correlation between protons and neutrons as there is in an electrically neutral atom. Figure 6.3.1 illustrates the particles of an atom for helium.

Radioactivity

As stated, in the nucleus of an atom with a given number of protons, nuclear stability exists with a specific number of neutrons. It is assumed that the neutrons provide stability in the nucleus by acting as a "nuclear cement" that counteracts the repulsive forces and increases the attractive forces of the positively charged protons confined in a small area. It is entirely possible, however, that an atom exists having a different number of

Figure 6.3.1

Components of the atom.



p = PROTON

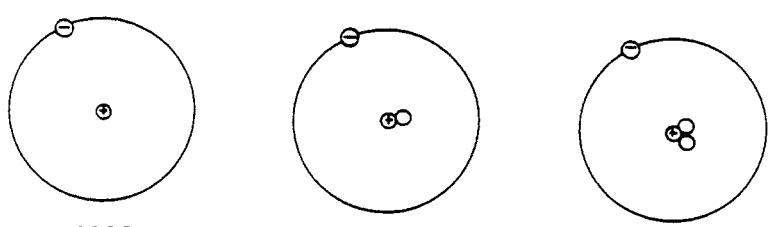
n = NEUTRON

e = ELECTRON

neutrons, other than the number that causes stability, for the same number of protons. When an element has the same number of protons but a different number of neutrons, the element with the excess neutrons is referred to as an isotope. Figure 6.3.2 shows the element hydrogen with its isotopes. In each of the atoms--hydrogen, deuterium, and tritium--the number of protons and electrons remains the same (one of each) and the number of neutrons varies. As shown, hydrogen is the stable atom; deuterium is a stable isotope of hydrogen; and tritium is a radioactive isotope of hydrogen.

Figure 6.3.2

Isotopes of hydrogen.



⊕ PROTON

⊖ ELECTRON

○ NEUTRON

With respect to nomenclature, the symbol for each of these isotopes is as follows:

Hydrogen: $_1H^1$
Deuterium: $_1H^2$--Hydrogen-2; ^2H
Tritium: $_1H^3$--Hydrogen-3; ^3H

In the first type of nomenclature, the subscript to the left of the element symbol indicates the number of protons in the element. The superscript to the right of the element symbol indicates the atomic weight of the isotope. Because the atomic weight attributed to the orbital electrons is usually considered insignificant (5.4×10^{-4} amu), the number of neutrons in the isotope being discussed can be calculated by subtracting the atomic weight from the number of protons. Therefore, using this approach, the number of neutrons in the isotope $_{92}U^{238}$ would be 146. The remaining two nomenclature forms merely list the element and atomic weight. These are important concepts because many elements have a number of isotopes, and these isotopes are present in the literature in any one of the preceding formats.

Because the isotope of an element has an improper combination of protons and neutrons, the nucleus of the atom is at a higher energy level and is said to be unstable. The term, radioactivity, refers to this improper combination of protons and neutrons that is in an unstable energy state. In an effort to find a more stable state, the atom will spontaneously transform, emitting some type of radiation to release energy. This is analogous to an orbital electron going from a higher energy orbital to a lower energy orbital and, in doing so, giving off light energy. In the instance of radiation being emitted from the nucleus, however, the radiation (energy) may be emitted as:

1. Alpha particle--A high-energy particle composed of two protons and two neutrons. It has a mass of 4 amu and a charge of +2.

2. Beta particles--Electrons emitted by the nuclei which may be positively or negatively charged. The mass of a beta particle is insignificant.

3. Gamma radiation--Bundles (photons) of electromagnetic radiation.

4. A combination of the above emissions.

Alpha Particles. The alpha particle originates in the nucleus of a radioactive atom. It is composed of two protons and two neutrons. It has a mass of four atomic mass units (amu) and a charge of +2. Because of its structure, it is the same as the nucleus of a helium atom.

Depending upon the radionuclide source, the energy of an alpha particle varies and may be as high as 10 million electron volts (MeV). Because of its energy, the alpha particle causes more ionization (formation of ions in an absorbing material) than a beta or gamma radiation would in an absorbing material. However, because of its large mass and positive charge, the distance the alpha particle can travel is very short. For example, the general range of an alpha particle in air is approximately 4 inches. It can

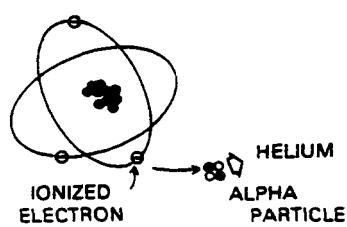
be stopped by a film of water, sheet of paper, or other paper-thin materials. Because of its short range, the danger in alpha particle radiation is not penetration through the skin but entrance of alpha particle emitters into the body through the respiratory or digestive systems.

The emission of alpha particles occurs through the radioactive decay of the larger elements. An example of an alpha source is Thorium-232 or Uranium-238. As discussed, if an alpha particle is emitted, the atomic weight and atomic number of the emitting radionuclide are altered. Referring to the sample radioactive decay (Figure 6.3.3), the alteration of the elements can be seen each time an alpha particle is emitted.

Once emitted, the alpha particle will attract two electrons to form a helium atom. Because the alpha particle causes the removal of two electrons from other atoms or molecules, the alpha particle is said to cause direct ionization (Figure 6.3.3).

Figure 6.3.3

Alpha particle interaction.



Beta Particles. The beta particle, as with the alpha particle, originates in the nucleus of a radioactive atom. The beta particle is an electron emitted during radioactive decay. The beta particle can be positively or negatively charged; but in either case, the atomic mass of the beta particle is negligible when considering the change in atomic weight of the radionuclide. The energy level associated with a beta particle is usually in the range of 0.017 to 4.0 million electron volts (MeV).

As previously stated, the beta particle may be positively or negatively charged. If the beta particle has a positive charge, it is referred to as a positron. In the formation of a positron, a proton in the nucleus assumes the properties of a neutron, and the beta particle is emitted with a positive charge. Because a proton is lost in the process, when a positron is emitted the atomic number of the element decreases by one. However, because only themass of an electron is emitted, the atomic mass of the element remains the same. For example, when the element Zinc-65 ($_{30}Zn^{65}$) emits a positron, the new element which would be produced because of the loss of a proton would be copper ($_{29}Cu^{65}$). Once the positron is emitted, it will collide or

interact with an electron in the medium and be annihilated. Upon annihilation, gamma radiation is emitted.

If the beta particle emitted is negative, then it is referred to as a negatron or high-speed electron. In the nucleus, a neutron assumes the basic properties of a proton, and a negatron is emitted. Because a proton is added to the nucleus, the atomic number increases by one, and a new element is formed. Once emitted, the high-speed electron will interact with the medium and cause direct ionization along with the formation of X-radiation. An example of beta decay that involves the emission of negatrons may be observed in the radioactive decay series (Figure 6.3.4). As shown in Table 6.3.1, each time a negatron is emitted, the atomic mass remains the same while the atomic number increases by one. Because each element is defined by its atomic number, each time a negatron is emitted, a new element is formed.

Table 6.3.1

Positron versus negatron.

Beta	Description	Charge	Mass (amu)	Energy Level	Effect of Emission
Positron	High-speed electron with positive charge	+1	0.00054	Up to several MeV	Z number decreases by 1
Negatron	High-speed electron with negative charge	-1	0.00054	Up to several MeV	Z number increases by 1

The penetration capability of the beta particle is somewhat greater than the alpha particle. This is due to the smaller mass and charge of the beta particle. For example, the average range of a beta particle in air is less than six feet; in wood, 1.5 inches; in human tissue, 0.1-0.5 inches. Beta particles can be stopped by material of low atomic weight; e.g., aluminum. In general, this is the type of shielding material used when dealing with beta radiation.

As with alpha radiation, beta radiation causes direct ionization. Further, as discussed, the interaction of the positron with the medium causes the formation of gamma radiation; and the interaction of a negatron with the medium causes the formation of X-radiation. Because of its relatively short range, the beta particle provides a minimum hazard. However, because of the potential formation of secondary gamma and X-radiation with somewhat longer range, the beta particle is a greater external radiation hazard than the alpha particle.

Gamma Radiation. Gamma radiation is a short-wave electromagnetic radiation. It originates from the nucleus; but emission of gamma radiation does not cause any change in the element's properties, atomic number, or atomic weight. Gamma radiation is emitted because the nucleus is in an unstable, excited state; and the emission of gamma radiation releases energy and causes the nucleus to fall to a more stable energy level.

The energy level associated with gamma radiation is the highest electromagnetic radiation that will be discussed. The energy level of gamma radiation is dependent upon the radionuclide source but usually falls within the range of 0.15 to 4 million electron volts. The source of gamma radiation is the radioactive decay of radioisotopes and the destruction of positrons as previously described.

Because gamma radiation has no mass or charge, is a wave, and has a relatively high energy content, gamma radiation has the potential for very deep penetration. This capability of deep penetration presents a tremendous health problem in the working environment. For example, to dissipate the energy of 1 millielectron volt of gamma radiation to one-half its original energy level, it would be necessary to have a shield of 0.5 inches of steel. Considering that potential energy levels of gamma radiation can reach one million times this value, gamma radiation is an obvious problem.

The photons associated with gamma radiation interact with atoms and molecules of the absorbing material in three ways.

1. Photoelectric effect--Incident photons cause the ejection of orbital electrons. These electrons possess energy equal to the difference between the photon energy and the electron binding energy. As shell vacancies are corrected, X-radiation is formed. Also, the resultant electron may interact to form X- or gamma radiation.
2. Compton effect--Incident photon gives up part of energy to orbital electrons. The electrons may recoil and be rejected. Further, the degraded photon may interact further with other electrons.

3. Pair production--High-energy photon interacts with electric field surrounding a charged particle (nucleus). The interaction causes the formation of an electron and positron of equal energy. When the positron and electron collide or slow down, X- or gamma radiation is formed.

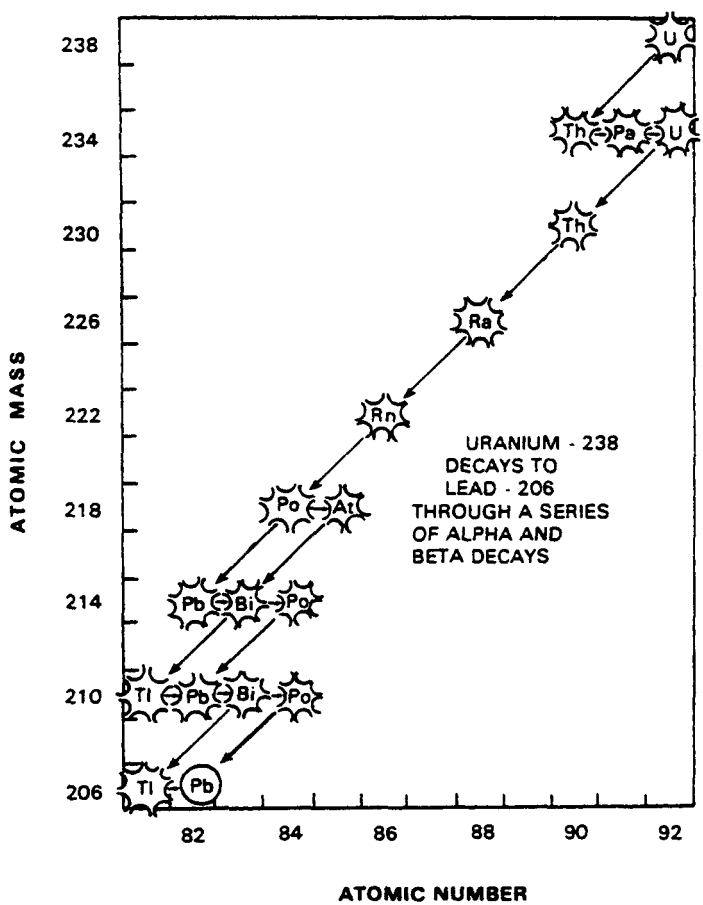
There are approximately 240 radionuclides (radioactive isotopes) which undergo this spontaneous transformation (radioactive decay) to reach a more stable energy level. Depending upon the type of emission, the isotope may undergo a change and, in essence, form a new element or new isotope. If, for example, an alpha or beta particle is emitted, a new element would be formed because protons are either gained or lost. If a neutron is emitted, the element is the same; but a new isotope is formed. If gamma radiation is emitted, then the element would remain the same; but the energy level in the nucleus would be lower.

It is entirely possible that, as radioactive decay occurs, the new isotope that is formed is also in an unstable state. It too would then undergo a spontaneous transformation, emit some type of radiation, and form a new element. Each of the new radionuclides formed from the spontaneous transformation is referred to as a "daughter product." Spontaneous transformation and the production of daughter products will continue until the chain of events forms a stable isotope or element. The decay change presented in Figure 6.3.4 illustrates the decay of Uranium-238 ($_{92}U^{238}$) to Lead-206 ($_{82}Pb^{206}$) through a series of alpha and beta emissions.

The steps in the decay process are shown in the diagram, where each nucleus that plays a part in the series is shown by a circle. The vertical

Figure 6.3.4

Uranium-238 decay.

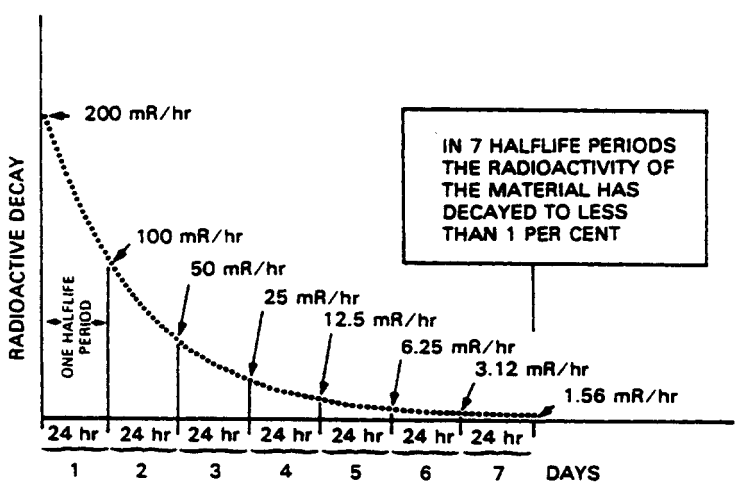


column containing the circle indicates its atomic number, and the horizontal column indicates the atomic mass. Each arrow that slants downward toward the left shows an alpha emission (two protons and two neutrons), and each arrow that points to the right shows a beta emission (electron). Notice that some of the nuclei in the series can decay in more than one direction. This is but one of four similar radioactive series that can occur in nature. Because of this chaining effect of radioactive decay, it is possible for the total radioactivity to increase beyond the radioactivity present in the original isotope.

The radioactive decay rate of a radionuclide is measured in terms of a characteristic time, the half-life. The half-life of a radioactive material is the time needed for half of the active atoms of any given quantity to decay. Isotope X, for example, has a half-life of 24 hours. This means that half of any given specimen of Isotope X will be converted to some other element by the end of 24 hours. Then, in the next 24 hours, half of the remaining Isotope X will decay, leaving only one-fourth of the original number of Isotope X atoms. Therefore, the number of atoms decaying is proportional to the number of atoms present. This is constant for any radionuclide. Figure 6.3.5 illustrates a typical half-life scheme ($T^{1/2}$) for an isotope having a half-life value of 24 hours. As can be seen by the illustration, by the end of the third half-life, the radioactivity has dropped to approximately one-eighth of its original activity; and by the end of the seventh half-life period, the activity has decreased to less than one percent of the original activity. Using this illustration, one can see that the longer the half-life value, the smaller the amount of radiation that will be released at any given time.

Figure 6.3.5

Decay of radioactive material.



There are two other types of ionization radiation which are important but are not formed by radioactive decay. The two forms of radiation to be discussed are X-radiation and neutrons.

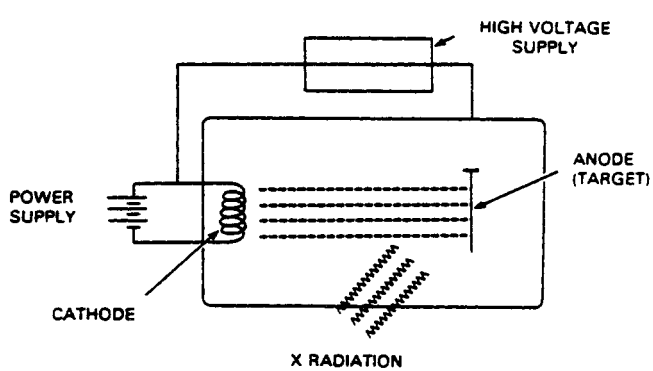
X-radiation. X-radiation is also electromagnetic radiation. It is similar in property to gamma radiation and interacts with absorbing materials in a similar manner, but it does not have as high an energy level as gamma radiation.

X-radiation originates outside the nucleus. In general, the principle behind the formation of X-radiation is that high-speed electrons (negatrons) are slowed down or stopped. In the process of slowing down, the electrons give up energy in the form of X-radiation. The quantity of energy released is dependent upon the speed of the electron and the characteristics of the medium (striking target). Generally, there are two types of radiation formed: bremstrahlung, which is caused by deflection of electrons traveling near a nucleus; and characteristic, which is produced in energetic transitions between orbital electron levels.

X-radiation is generally produced by machine, although some X-radiation is emitted from the collision of negatrons during the process of radioactive decay. The machine necessary to produce X-radiation has basically three components (Figure 6.3.6): (1) cathode (electron source), (2) anode (target), and (3) potential difference between the cathode and the anode in a vacuum. In the machine, a potential difference is established between the cathode and the anode. The electrons travel through the vacuum at high speeds until they strike the target (anode). The energy released from their collision is in the form of X-radiation.

Figure 6.3.6

X-ray machine.



The penetrating capability of X-radiation is dependent upon the wavelength of the radiation. X-radiation with a short wavelength (approximately 0.001-0.1 nm in length) is called "hard" X-radiation. This type of radiation has the capability to penetrate several inches of steel. X-radiation with longer wavelength (0.1-1.0 nm) is called "soft" and is less penetrating than hard X-radiation. The range of penetration of X-radiation is dependent upon the energy of the radiation and the type of medium it is penetrating. For example, the half-value layer (thickness required to reduce incident radiation by 0.5) is several inches of concrete. As previously discussed, the interaction with the target medium is the same for X-radiation as with gamma radiation.

Neutrons. The final type of ionizing radiation to be discussed is the neutron. The neutron particle originates from the nucleus. It has an atomic mass of 1 amu and a charge of 0. The energy level usually associated with neutrons is greater than gamma or X-radiation but is less than alpha or beta radiation. However, because of the zero charge on the neutron, the neutron has a greater penetrating capability than the alpha or beta particle.

Neutrons are formed in basically two ways. First, it is possible to bombard a light element, e.g., beryllium or lithium, with alpha particles or gamma radiation and increase the energy in the nucleus so that a neutron will be emitted. The second source is the fission of isotopes of uranium or plutonium. This fission is caused by neutron bombardment and results in the production of more neutrons. Because the reaction is produced by neutrons and results in the production of neutrons, a chain reaction may be established such that the neutrons resulting in the fission of a given uranium or plutonium isotope cause the fission of more isotopes. Because of this, the principle of criticality evolves. There is a minimum level of concentration required of the radioisotope to allow the chain reaction to begin. This is usually referred to as the "critical level." If the critical level is not exceeded, the chain reaction will not occur. If, however, the critical level is exceeded, a chain reaction may be produced by the fission of a single plutonium atom.

Neutrons are developed through nuclear reactions as described above or are mechanically produced through the use of accelerators such as the Van de Graff or the Cockroft-Walton generator.

The interaction of neutrons with a target medium is dependent upon the energy level of the neutron. Neutrons of a high energy level (fast neutrons) collide with the nuclei of the absorbing material and lose energy in a billiard-ball-like collision. Neutrons with less energy (slow/thermal neutrons) are captured by the absorbing nuclei. In either case, alpha, beta, or gamma radiation is emitted. Because ionization is secondary, the calculation of neutron dose is difficult.

The penetrating capability of the neutron is dependent upon the energy level, characteristics of the medium, and type of collision which occurs. The average distance the neutron of a given energy level will travel before some type of interaction is referred to as mean free path (mfp). The probability of an interaction in three mean free paths is equal to 0.95. As an example,

in human tissue, the mean free path of a neutron may vary from 0.25 to several inches.

Table 6.3.2 summarizes the ionizing radiation discussed, and Figure 6.3.7 illustrates the relative penetrating capability of the types of ionizing radiation.

Units of Measure

When discussing radioactivity and the energies involved, the following units of measure will be of value. When reviewing the terms, place emphasis on the concepts of each term rather than the actual numerical value.

Units of Energy. In physics, the standard unit of energy is the joule. The joule represents the work done when a constant force of one newton moves a body a distance of one meter in the direction of the force. Energies of atomic and nuclear phenomena are usually given in terms of electron volts (eV). One electron volt is the kinetic energy acquired by an electron after being accelerated through a potential difference of one volt. Energies encountered in radioactive decay processes are usually expressed in terms of thousands of electron volts (KeV) or millions of electron volts (MeV). For example, the energy of Cesium-137 gamma radiation is 0.667 million electron volts. If necessary, it is possible to convert joules to electron volts using the factor:

$$1{,}000{,}000 \text{ electron volts (eV)} = 1.602 \times 10^{-13} \text{ joules (J)}$$

Units of Activity. The measure of the rate of radioactive decay is given in terms of activity. The activity reflects the number of radioactive atoms emitting ionizing radiation. The special unit of activity is the curie (Ci) and is defined in terms of events per unit time, disintegrations per minute, or counts per minute. The numerical value for a curie is as follows:

$$1 \text{ curie (Ci)} = 3.7 \times 10^{10} \text{ disintegrations per second}$$

When discussing a quantity of radioisotopes, the quantity is discussed with respect to the amount of activity of that quantity. Therefore, it would be more appropriate to state that a quantity of uranium is "an amount of uranium with an activity of 500 curies" rather than "500 curies of uranium."

Exposure. The roentgen (R) is the special unit of exposure. This unit refers specifically to the measure of the amount of charge produced in air by gamma or X-radiation. Numerically, one roentgen corresponds to the production of about 2×10^9 ion pairs per centimeter^3 of dry air at standard temperature and pressure. Exposure rate is the time derivative of exposure and is usually expressed in terms of roentgens per hour, roentgens per minute, or milliroentgens per hour, etc. The exposure rate constant (Γ) is the exposure rate per curie of radioisotope at 1 meter. The values of Γ are tabulated in the literature for many gamma emitters.

Absorbed Dose. The absorbed dose (D) is the energy imparted by ionizing radiation to absorbing matter per unit mass of absorbing material. The

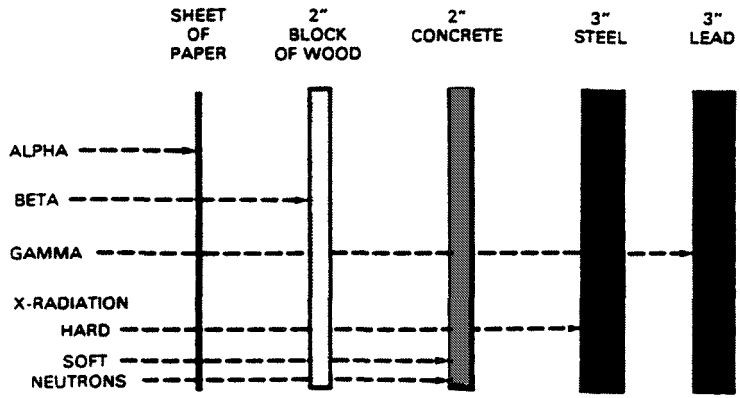
Table 6.3.2

Properties of ionizing radiation.

Type	Source	Description	Charge	Mass (amu)	Energy	Radiation Hazard	Necessary Shielding
Alpha	Natural and manmade isotopes	Helium nucleus--causes direct ionization	+2	4	2-9 MeV	Internal hazard only	Thin sheet of paper
Beta	Most radioactive isotopes	Electron (+ or -) ionizes 1/100 as heavily as alpha	± 1	.00054	Up to several MeV	Internal or external danger	Thick sheet of cardboard
Gamma	Natural isotopes-nuclear reactions	High energy electromagnetic radiation	0	0	Up to several MeV	Dangerous--deep penetration	Heavy shielding -- e.g., inches of lead
X	Slowing of electrons	High energy electromagnetic radiation--less than gamma	0	0	Up to several MeV	Dangerous--deep penetration	Heavy shielding
Neutrons	Nuclear reactions	Uncharged particles	0	1	Up to several MeV	Hazardous	Thick shielding composed of light elements

Figure 6.3.7

Ionizing penetration.



standard unit of absorbed dose is the rad and is defined as 1 rad = 10^{-2} joules/Kg. The absorbed dose rate is the time derivative of absorbed dose and reflects the given absorbed dose over a given period of time. It is usually expressed in terms of millirads per hour, rads per second, etc.

Dose Equivalent. Because of the specific biological effects and actions of radiation upon human tissue, the measure of exposure or even absorbed dose does not present completely the measurement of ionizing radiation. For example, each type of ionizing radiation has a different penetrating capability. Alpha particles will have no detrimental effect to the body unless they enter the body through some means other than penetrating through the skin; whereas, gamma radiation easily penetrates the protective layer of the skin and can cause severe internal damage. Further, certain radionuclide sources, if introduced to the body, will migrate to certain areas of the body. An example of this is the radioisotope of calcium. Because of their properties and involvement in the development of bone tissue, most radioisotopes with properties similar to that of calcium will tend to concentrate in the body in the bone areas. Because of this, one cannot evaluate the problem of radiation based upon a whole-body exposure. Finally, the effect of the various types of radiation on different parts of the body will vary. For example, areas of the hands, arms, and feet are less sensitive to radiation exposure than other areas such as the eyes, internal organs, or the reproductive organs.

To deal with these variations, a standard unit of dose equivalent has been developed. It attempts to take into consideration the above-mentioned factors. Generally, the unit, known as the rem, is equivalent to the exposure (R) or the absorbed dose (rad) multiplied by the quality factor. Although not absolute, it will be assumed that this quality factor (QF) corrects for the

relative biological efficiency of the different types of radiation (RBE). The exact QF varies according to the source of radionuclide, area of exposure, and type of ionizing radiation. However, to calculate the dose equivalent, the following values may be used for purposes of determining the rem value, given exposure or absorbed dose.

Calculation of dose equivalent.
[rem = rad (or R) x QF]

Type	QF
X-radiation	1.0
Gamma radiation	1.0
Beta radiation	1.0
Alpha radiation	20.0
Neutron (fast)	3.0
Neutron (slow)	10.0
Fission fragments	20.0

An example of the calculation is as follows:

Example

If the absorbed dose reading from a $_{88}Ra^{226}$ (alpha emitter) is found to be 0.05 mrad/hr at a distance of 1 meter, what is the dose equivalent for an exposure lasting 8 hours?

Solution

$$\text{rem} = \text{rad} \times QF$$

$$= \frac{0.05 \text{ mrad}}{\text{hr}} \times \frac{20 \text{ mrem}}{\text{mrad}}$$

$$= \frac{1.0 \text{ mrem}}{\text{hr}}$$

$$\text{rem/8 hr} = \frac{1.0 \text{ mrem}}{\text{hr}} \times 8 \text{ hr} = 8.0 \text{ mrem}$$

Although the quality factor (QF) is not exact, it can be used to calculate the dose equivalent given the exposure or absorbed dose rate for low level sources. In later sections, the importance of dose equivalent will be presented.

Fluence. When evaluating particulate radiation (alpha, beta, neutron) it may be useful to know the number of particles entering a cross section of a given area. The unit used to express this value is fluence (Φ) and is expressed in terms of "particles per centimeter^2." If the sum of the energies of the particles that enter the cross sectional unit is to be discussed, then the unit is energy fluence (ψ) and is expressed in terms of "millielectron volts per centimeter^2."

Flux Density. If the fluence is to be determined over a given period of time and rate calculated, then this rate of fluence per unit time is referred to as flux density (ϕ). This is usually expressed in terms of "particles per centimeter^2 per second"; e.g., neutrons per centimeter^2 per second. The energy flux density is the energy fluence of a given period of time and is expressed in million electron volts per centimeter^2 per second.

Each of the terms presented in this section will be used throughout the remainder of the sections on ionizing radiation. It is not necessary to memorize all numerical values, but it is important to understand the concept behind each unit of measurement.

Biological Effects of Ionizing Radiation

The fundamental property of ionizing radiation relates to the transfer of energy when the radiation passes through material. Absorption of the energy may cause ionization (the expulsion of orbital electrons from the atom or molecule) or excitation of the orbital electrons to a higher energy state. The ions formed in turn react with other atoms and molecules in the absorbing material, causing potential changes in those molecules and the absorbing structure. Because ionizing radiation causes changes at the atomic and molecular level, the effects of ionizing radiation to the cells will be discussed initially; and then the overall systems function will be discussed.

The energy required to cause cell death is very small. A lethal dose of radiation dissipates approximately one-millionth the amount of energy dissipated by the cell during its normal daily function. Thus, a lethal dose affects only one of 2.0×10^7 molecules in a cell.

In the cell, ionizing radiation has two actions. First, if the ionizing radiation irradiates critical molecules of the cell, e.g., DNA or mitochondria, immediate cell death will occur. An indirect action can also occur by the irradiation of water molecules. This irradiation forms active products, e.g., H_2O_2, OH^-, O_2H^-, which react with other molecules within a cell and cause damage or death. Because irradiation is random at the cell level, the indirect action is more significant than the direct action. In general, radiation causes the following: immediate cell death, cell damage that prevents growth or causes the formation of cell mutations, or reduction of cell function and ultimately body function.

All forms of ionizing radiation produce some type of injury to the cell. Tissue reaction is dependent upon the density of the ionization in the radiation path; i.e., linear energy transfer. Particulate radiation (alpha,

beta) produces more damage per energy absorbed, and thus a high relative biological efficiency and a greater linear energy transfer factor (amount of energy transferred per linear penetration). Electromagnetic radiation, because of its high penetration capability, causes more diffuse ionization. It should be noted that the particulate radiation is more damaging if allowed to enter the body (internal radiation hazard). Because particulate radiation has a small penetrating capability, the protective layers of the skin tend to provide a barrier to prevent the penetration of particulate radiation through the skin.

The response of an individual to radiation is dependent upon the dosage of the radiation; the amount and type of tissue irradiated, e.g., localized versus whole-body irradiation; and the length of time of exposure. A localized exposure; e.g., to the hands and arms, will cause damage to the specific irradiated site or organ. However, it is possible for a local exposure to cause systemic changes. Whole-body irradiations tend to cause systemic-type illnesses. Signs and symptoms such as nausea, vomiting, skin erythema, intestinal bleeding, and diarrhea are common to whole-body irradiation. The following effects would be observed in man after acute whole-body doses of penetrating radiation:

1. 0-25 rad--no observable effect.

2. 25-50 rad--minor temporary blood changes.

3. 50-150 rad--possible nausea and vomiting, along with reduced white blood cell count.

4. 150-300 rad--exaggeration of above symptoms plus diarrhea, malaise, and loss of appetite.

5. 300-500 rad--exaggeration of above symptoms, plus hemorrhaging and loss of hair (depilation). About 50 percent of the untreated exposed population will die at 450-500 rad level.

6. Above 500 rad--Most of the above symptoms will appear sooner in more severe form. Survival chances diminish rapidly with higher doses.

Acute high-level doses or moderate doses over longer periods of time can produce effects later in life. The most notable effects include increased risk of cancer or leukemia; nonspecific life span shortening, i.e., acceleration of aging process; and harmful mutations that may be transmitted to future generations. In general, exposure to ionizing radiation causes a decrease in the efficiency of cell activity to the point that cell function is diminished and may ultimately cause cell death. To minimize the effect of ionizing radiation in the occupational environment, maximum levels of ionizing radiation exposure have been established.

Industrial Uses of Ionizing Radiation

For many years following the basic investigations in the field of radioactivity, the primary use of ionizing radiation was in the medical field;

and little progress was made in integrating ionizing radiation in industrial areas. Recent advances, such as fission and fusion techniques, have brought about the availability of economic radionuclides that are rapidly being introduced into the industrial field.

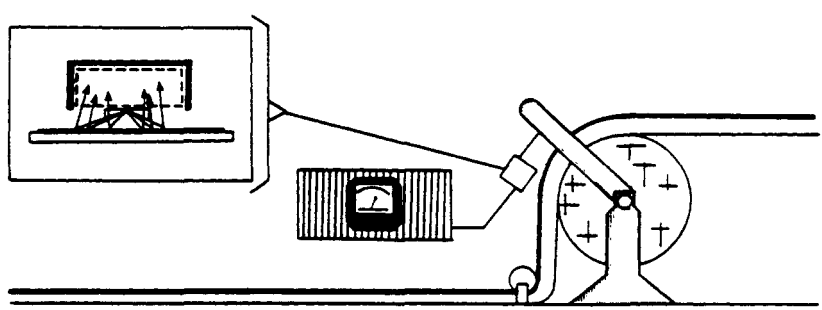
In most countries, only a small percentage of the total work force is involved in the industrial use of ionizing radiation. With few exceptions, the use of ionizing radiation has been remarkably free of radiation accident and injury. However, because of the potential hazard, much consideration must be given to the health and safety aspects as well as to the actual industrial applications of ionizing radiation. The following are examples of uses of ionizing radiation in the industrial environment.

Radiation Gauges. Ionizing radiation penetrates or is reflected by matter; and, with a suitable detector, it is possible to extract useful information from the transmitted or reflected beam. Because the intensity of radiation used is small, the beam is nondestructive and lends itself to automation. Some uses of a radiation gauge include the determination of sheet metal thickness, the density of metals or fluids, or the calculation of moisture content.

There are two types of gauges in use. The first is a transmission gauge that monitors the radiation penetrating a given substance. When using this type of gauge, the factors that will vary the transmission reading include the density and composition of the target material. The second type of gauge is a reflection-type gauge. It is more sensitive and in many instances may be the only practical solution. An illustration of a "backscatter" or reflection-type radiation gauge is given in Figure 6.3.8.

Figure 6.3.8

Reflection-type thickness gauge.



In these gauges, beta emitters are used in conjunction with a reflecting or backing material such that the backscatter electron intensity is linearly proportional to the thickness of the substance to be checked over a given range. These gauges are finding wide application because of the increasing industrial importance of thin films and coatings.

If the gauges automatically control the process, e.g., metal sheet rolling, then a greater accuracy can be obtained than with conventional control methods. Table 6.3.3 illustrates the increase in quality control when using a radiation gauge with automatic control versus a conventional type control sampling method.

Table 6.3.3

Process quality control comparison.

Industry	Application	Process Control (95% Correlation) Conventional Control (Normal sampling method)	Radiation Gauge With Manual Control	Radiation Gauge With Automatic Control
Paper	Fourdrinier paper machine	±12%	±10%	±3%
Metals	Tandem cold roll mill	±10%	±7%	±1.5%
Rubber	Sheeting calender	±10%	±6%	±3.2%
Rubber	Tire fabric calender	±6.5%	±5.5%	±3.5%
Plastics	Calender	±7%	±3.5%	±2.5%
Impregnation	Saturation dip coater	Automatic control provided an improvement factor of 2.5 over conventional		
Abrasive	Abrasive maker	Automatic control provided an improvement factor of 2 over conventional		

(From Crompton, C. E.)

The three types of ionizing particles in common use in radiation gauges are photons (X or gamma), beta (normally negative electrons), and neutrons (fast and thermal). A common source for industrial radiation gauge is a sealed source of Strontium-90 of up to 1 curie activity. All radiation sources made for such use must meet rigid specifications regarding resistance to temperature, pressure, impact, vibration, and puncture. Safety experience

with such sources has been favorable; even gauges exposed to intense heat and fires have evidenced little or no leakage. Their locations and layout are normally such that external exposure of personnel is minimal. However, all sources should be periodically checked for leakage and maintained by qualified personnel.

Radiography and Fluoroscopy. Radiography is defined as the production of a shadow image of the internal structure of an object on a permanent record, usually by some type of film and template. The primary types of ionizing radiation used include photons (X or gamma) or neutrons (fast or thermal).

Radiography provides a close inspection during and after fabrication of materials or structure. Welds in nuclear power plants or high pressure pipelines, critical joints in the foundation of large buildings, and engines in large jet planes are only a few examples in which such a radiographic section plays a critical part. Industrial radiography comprises both field radiography and stationary radiography. Both are based on the same principle; but in field radiography the source of radiation is taken to the specimen, while in stationary radiography the specimen is taken to the source of radiation.

Fluoroscopy works on the same principle as radiography with the exception that the shadow image is presented temporarily on a screen. It is most appropriate for use in industry as a quality control tool for mass produced components, such as turbine blades and transistors, and for the inspection of such items as cables. Fluoroscopy is about four times less sensitive than radiography; therefore, it can be used only where such low sensitivity can be tolerated and is usually backed up by radiography.

In either of the above techniques, the primary hazard is the radiation source that is being used as part of the procedure.

X-Ray Diffraction and Fluorescent Analysis. It has been known for many years that the planes in a molecular crystal lattice will diffract X-radiation in a set three-dimensional pattern. Using this information, if X-radiation is emitted through a molecular structure and the diffraction pattern determined, it is then possible to evaluate the crystal state of the material. Further, it is also possible to determine the actual content of the material based upon the absorption and emission properties of ionizing radiation of that material.

The procedure of crystals for broadcast transmitters, watch movements, silicone and germanium semiconductors are all aided by the determination of crystal structure through X-ray diffraction techniques. The determination of crystalline structure of a finished product, such as a turbine blade or nuclear reactor vessel, provides information on the soundness and freedom from stress which are important in predicting the useful life of the product.

The personnel hazards are of a specialized nature. The greatest hazard is associated with very fine, intense X-ray beams. Further, the electron accelerating voltage is normally in the range of 30,000 to 50,000 volts. It is possible, therefore, that the primary beam of X-radiation may have an intensity as high as 7000 roentgens per second; and any part of the body held

in the beam for only fractions of a second would receive such a dose. Besides the risk of exposure to the primary beam, a hazard also exists of exposure to a beam scatter off the specimen or analyzing crystal, which could be quite intense. Much less intense, but still strong enough to make further shielding precautions advisable, is the scattered radiation around joints, around shielding, and in general when the radiation is scattered at least twice.

The prime requisite for the safe operation of an X-ray diffraction and fluorescent unit consists of proper training and supervision of operation personnel. Accidents frequently occur when operators place their fingers and hands in the primary beam to remove samples, forgetting that the beam is on.

Electron-Beam Equipment. The development of reliable electron guns and electron optics using electric and magnetic fields has led to many interesting and important industrial applications of electron beams. Two of them, electron-beam evaporators and electron-beam welders, take advantages of the unique ability of a defined electron beam to introduce a large amount of heat at a specified point or area in a specimen. Two other applications, electron microscopes and scanning electron microscopes, take advantage of the fact that the electron beam can behave like a light wave and is capable of an extremely high resolving power.

The electron-beam evaporator uses the heat generated when an electron beam from a cathode strikes a target after being accelerated by application of a high-potential, concentrated biomagnetic or electric field acting as an electromagnetic "lens." The power emitted by the electron beam is absorbed as heat at the surface of the target, causing the target surface to evaporate at a rate that may be controlled by varying the voltage and the current of the electron source. Metals such as tungsten and molybdenum and nonmetals such as quartz and aluminum oxide may be evaporated in this way. The evaporated metal is then used to provide thin antistatic coatings on glass or plastic and, in general, to meet other needs in the field of solid-state and laser technology. The hazards normally associated with electron-beam evaporators are the production of X-radiation, the high energy levels of electron beams, and the high voltage required to accelerate the electrons.

Electron-beam welders operate much like electron-beam evaporators using higher voltages and tighter focusing to concentrate the beam at a point on the target. The advantage of electron-beam welding versus conventional welding is that the work piece is heated from the inside, thereby eliminating the distortion often associated with conventional welding techniques. The hazards associated with the electron-beam welder are similar to those of the electron-beam evaporator with the added hazard of higher voltages; those required by the welder are in the range of 150,000 volts rather than the 13,000 to 20,000 volts required by the evaporator.

Electron microscopes are used in research and industrial applications when resolving power approaching the ultimate possible is required. An optical microscope is limited by the wavelength of visible light which may be taken as 500 cm. In the electron microscope, however, the theoretical limit is a function of the electron beam accelerating voltage.

The basic components of an electron microscope are the electron gun, composed of a cathode and anode, and the condensing magnetic lens, specimen state, objective magnetic lens, projection magnetic lens, and viewing screen. The electron beam from the cathode is accelerated to the anode and concentrated into a parallel beam by the condensing lens. The beam then passes through the specimen. The energy of the electrons must be sufficient to penetrate the specimen, and accelerating voltages of 50,000 to 60,000 are commonly used. Passing through the objective lens, a magnified image is formed at the intermediate lens which converges the electron beam before passing through the final magnification stage. The final image can usually be viewed through observation ports on a fluorescent screen. Magnification as high as 200,000 times and a point resolution of 0.4 nanometers are now available on commercial models.

As with any of the electron beam devices, the hazards associated with the electron microscope are similar. The unit should be monitored for leakage of X-radiation, and precautions should be taken when dealing with the high voltage of the unit.

Activation Analysis. In this technique of chemical analysis, the test material is irradiated by one of several different types of uncharged or charged particles. A probability exists that in the reaction the target nuclei will be transformed into unstable or radioactive nuclei, the number of which depends upon the number of bombarding projectiles, the number of target nuclei, and the cross section of the reaction. Once radioactive, the target material will begin emitting ionizing radiation. Each radionuclide has a distinct pattern of X- and gamma radiation emission. Once an analysis of the emission is performed, then the identity and quantity of the elements present may be determined.

Radioactive Tracers. Isotopes have the same chemical properties whether they are radioactive or not. Therefore, it is possible to substitute a known quantity of radioactive isotope in a given material or process and then "trace" its progress. An example of the use of these tracers in product improvement is the study of the effect of various lubricating oil combinations on the wear of pistons in engines and cylinders. In this application, a piston ring was irradiated and was used on a piston in a cylinder of an internal combustion engine. The amount of activity in the lubricant was determined by counting and the location of wear was determined by auto-radiography. This study permitted the development of improved lubricants in a relatively short time.

Another example of the use of tracers is to determine wetting, detergency, absorption, and durability. To improve the accuracy of a radiochemical separation, one can add a known quantity of radionuclide to a mixture with an unknown quantity of a radionuclide of the same element, for example Strontium-85 to a Strontium-90 mixture, and determine quantitatively the amount of Strontium-90 in the mixture.

Still another example of the use of tracers is in determining the volume of complex reservoirs in hydrology by utilizing a technique known as "isotope dilution." A known volume of radioisotope at a known concentration is added

to a complex mixture or reservoir. If one assumes a perfect mixture and a sample is taken, it is possible to determine the volume of the reservoir by taking a known volume of sample and measuring its isotope concentration. From this, the actual volume of the entire reservoir or complex mixture may be extrapolated.

The tagging of a product in transit is an example of the use of radioactive tracers in the petroleum industry. In one application, various radioactive tracers are used to tag the input water in water flooding projects. Finally, in the field of solid-state technology, minute concentrations of metals and nonmetals can have profound effects on a semiconductor, and radioactive tracers are used as the only practical method of identifying behavior of trace elements found in reagents used in semiconductor fabrication.

Aerosol Fire Detectors. One of the most sensitive detectors of combustible gases and smoke relies on the use of radioactive materials. In this system, two ionization chambers, each containing a radioactive foil, are used. One is sealed and the other is open to the ambient air. These two chambers are connected to a cold-cathode, gas-discharge tube. In the absence of air contaminants, the gas discharge tube does not fire. If the air becomes contaminated with combustible gases and smoke particles, the tube fires, actuating an electromagnetic relay which sounds the alarm. The hazard associated with this device is minimal because of the size of the sealed source.

Luminescent Dials. One of the oldest applications of ionizing radiation involves the use of radium as a source of radiation in a self-luminous compound. If a compound of phosphor (zinc sulfide) is bombarded with ionizing particles, the phosphor scintillates, causing it to "glow." The hazards associated with self-luminous compounds are present in all stages of the process, including the preparation of the compound, painting and assembly, and inspection. In each stage, steps must be taken to minimize personnel contact.

Large Radiation Sources. The use of large radiation sources has achieved practicality with the advent of inexpensive radionuclides and engineering design advances in accelerators. One application of large radiation sources is radiation processing. This is the use of ionizing radiation to produce a biological or chemical change. Routine uses of radiation processing in industry include sterilization of medical supplies, an enzyme for laundry use, synthesis of ethyl bromide, controlled degradation of polyethylene oxide polymer, irradiation of wood-plastic composites, and control of anthrax bacillus in sheep hides and wood. Table 6.3.4 indicates industrial applications of radiation processing.

The second use of large radiation sources is a thermoelectric generator. The ionizing radiation is used as a heat source, and thermoelectric converters to convert the heat to electricity. The primary hazards involved are the presence of gamma and X-radiation, and neutrons produced by the procedure.

A third potential use of large source radiation is in nuclear explosions. They may be used in the construction industry for excavating and tunneling, in

the mining industry for breaking up ore, and in the petroleum industry for stimulation of gas fields. Although most industrial uses of radionuclides emphasize the safety and health protection of the working personnel, when using large radiation sources, protection of the general public must also be considered.

Agricultural Uses

Agricultural uses of ionizing radiation may be categorized into two groups. The first use is the irradiation of living tissue with the intent of modifying the matter. Examples of this use include irradiation of potatoes to inhibit sprout production, the sterilization of fruit, or irradiation of seeds to cause mutations in an effort to develop various strains of the plant. With these uses, machine sources as well as naturally occurring radioisotopes are used.

Widely used radioisotope sources in agriculture and research include:

1. Multikilocurie gamma sources for food disinfection, pasteurization, and sterilization.

2. Beta and less intensive gamma sources used in radiography, thickness and density and level gauges.

3. Alpha sources used in electrostatic discharges, gauges, and electronic warning devices.

4. Isotope accelerators used for neutron activation analysis of soil, metals, and protein.

As in industrial uses, tracers are used to evaluate biological phenomena. All types of radiation are used in agriculture. Alpha particles are used to study cellular disposition. Beta particles are used in evaluation of metabolic processes, pathways of fertilizer and nutrient deposition; and finally, X- and gamma radiation are used to evaluate processes similar to the use of beta particles.

Medical Uses

Medical uses for ionizing radiation may also be grouped into two categories. The first category is the application of X-radiation. Most people are familiar with its common use in the development of a film outline of bones, teeth, and calcified structures, along with the internal organs. Further, injection or ingestion of X-radiation opaque substance into the body provides outlines of the desired organ. For example, barium sulfate is ingested to provide a film of the intestinal tract.

The medical profession has also found various uses of radionuclides, primarily in diagnosis and therapy. When the radioactive isotopes of elements

that are normally found in various areas of the body are introduced via injection or ingestion, it is possible to chart the path and deposition of specific elements and compounds throughout the body. For example, if an analysis of the activity of the thyroid gland were of interest, then radioactive iodine would be used since iodine tends to migrate to the thyroid gland; if the kidney were of interest, radioactive mercury migrates to the kidney just as iodine does to the thyroid gland. Certain types of cancers may

Table 6.3.4

Some applications of ionizing radiation in industry.

A = Research B = Pilot Plants C = Production	Industrial Process*												
	1	2	3	4	5	6	7	8	9	10	11	12	13
Sewage Treatment	A												
Semiconductors													A
Flooring							A	B		A		A	
Furniture							A	B		A		A	
Hides and Hair	C		C	C	C								
Textile							B	A		B	A	B	
Adhesives							A	A			A		A
Rubber							A	A		B	B		A
Spices		A	A	A	A								
Paints and Coatings								B		B	A	A	A
Membranes	A						A	A		B		B	
Chemical Synthesis								A	A	B			
Fuels									A				A
Lubricants									A				A
Wood-Plastic Composites							A	B		A		A	
Plastic Piping							A	A		A		A	
Heat Shrinking Tubing							B	B		A			
Enzymes		A	A		A								
Cosmetics	B	A	A		A								
Pharmaceutical	A	A	A		A								
Medical Supplies	C				A								
Foods	B	A	A	B	A	B							

* 1 = Sterilization
2 = Pateurization
3 = Salmonella Control
4 = Insect Disinfestation
5 = Disinfection
6 = Sprout Inhibition
7 = Cross Linking
8 = Polymerization
9 = Scission
10 = Free Radicals
11 = Curing
12 = Grafting
13 = Testing and Evaluation

Source: Gamma Industries, Inc.

be identified by noting that the cancerous cells do not absorb or use the same elements of the body as do normal, healthy tissues. Therefore, if a radioactive isotope were introduced into a given area and the cells did not absorb the radioisotope as would be expected, then the location of the cancer could be identified.

The uses of radionuclides in therapy are rather well known. Because of the biological effects of ionizing radiation, it is possible to expose nondesirable (cancerous) cells to specific types of ionizing radiation that will not affect normal tissue; and the effect of the ionizing radiation will be to cause cell damage and death to the cancerous tissues.

Hazards. In general, each of the industrial, agricultural, or medical uses has basically the same hazard. In those instances where a naturally occurring radioisotope is being used, specific exposures to that isotope is the primary problem. In those instances where high voltage equipment is needed to create the radiation, e.g. X-radiation, then an alternate problem of potential high-voltage danger should be considered. Radioisotopes may be used for specific types of radiation, e.g., alpha source, but the isotope selected may also emit gamma or beta radiation. Care must be taken to evaluate properly the extraneous as well as desired radiation.

Maximum Permissible Dose

In an effort to minimize the effect of ionizing radiation exposure, the following maximum permissible dose (MPD) levels have been established. The dose levels are presented in units of dose equivalents (rem).

Using the MPD values presented in Table 6.3.5, it will be possible to calculate the exposure levels in the working environment that will not exceed the MPD level.

Efforts have also been made to convert the equivalent dose maxima to actual concentrations of radionuclides that may be present in the working environment, specifically in the air and water. Handbook #69, "Maximum Permissible Body Burdens and Maximum Permissible Concentrations of Radionuclides in Air and Water for Occupational Exposure," specifically identifies the maximum permissible burdens and permissible concentrations for each of the radionuclides presently being used in the working environment. The handbook also identifies the critical organ affected by that radionuclide. In essence, the handbook has converted the maximum permissible dose equivalent to a concentration of a specific radionuclide based upon the energy levels of the emissions of the nuclide. Table 6.3.6 is an excerpt from Handbook #69.

Table 6.3.5

MPD equivalent recommendations.

	Maximum Weekly Dose (rem)	Maximum 13-Week Dose (rem)	Maximum Yearly Dose (rem)	Maximum Accumulative Dose (rem)
Occupational Exposure				
Whole Body	0.23	3.0	12.0	5(N-18)*
Skin		10.0	30.0	
Hands, feet		25.0	75.0	
Forearms, ankles		10.0	30.0	
Non-Occupational Exposure	0.01			
Emergency Situation				
Whole Body	[illegible]0			
Hands and forearms	[illegible]0 additional			

*Assume personnel of age greater than 18 yrs.

Table 6.3.6

Maximum permissible body burdens and maximum permissible concentrations of radionuclides in air and in water for occupational exposure.

Radionuclide and type of decay		Organ of reference (critical organ in boldface)	Maximum permissible burden in total body $q(\mu c)$	Maximum permissible concentrations			
				For 40 hour week		For 168 hour week	
				$(MPC)_w$ $\mu c/cc$	$(MPC)_a$ μc cc	$(MPC)_w$ $\mu c/cc$	$(MPC)_a$ $\mu c/cc$
	(Insol)	**GI (LLI)**		2×10^{-3}	4×10^{-7}	8×10^{-4}	10^{-7}
		Lung			2×10^{-6}		6×10^{-7}
$_{34}Se^{75}(\epsilon, \gamma)$		**Kidney**	90	9×10^{-3}	10^{-6}	3×10^{-3}	4×10^{-7}
		Total Body	100	0.01	10^{-6}	3×10^{-3}	5×10^{-7}
	(Sol)	Liver	100	0.01	2×10^{-6}	4×10^{-3}	5×10^{-7}
		Spleen	200	0.02	3×10^{-6}	8×10^{-3}	10^{-6}
		GI (LLI)		0.07	2×10^{-5}	0.03	6×10^{-6}
		Lung			10^{-7}		4×10^{-7}
	(Insol)	GI (LLI)		8×10^{-3}	10^{-6}	3×10^{-3}	5×10^{-7}
$_{35}Br^{82}(\beta^-, \gamma)$		**Total Body**	10	8×10^{-3}	10^{-6}	3×10^{-3}	4×10^{-7}
	(Sol)	GI (SI)		8×10^{-3}	2×10^{-6}	3×10^{-3}	6×10^{-7}
		GI (LLI)		10^{-3}	2×10^{-7}	4×10^{-4}	6×10^{-6}
	(Insol)	Lung			6×10^{-7}		2×10^{-7}
$_{36}Kr^{85m}(\beta^-, \gamma)$	(Immersion)	**Total Body**			6×10^{-6}		10^{-6}
$_{36}Kr^{85}(\beta^-)$	(Immersion)	**Total Body**			10^{-5}		3×10^{-6}
$_{36}Kr^{87}(\beta^-, \gamma)$	(Immersion)	**Total Body**			10^{-6}		2×10^{-7}

Summary

In this chapter, the atom, radioactivity, and the process of radioactive decay have been briefly discussed. Five types of ionizing radiation, including alpha, beta, gamma, X-, and neutron have been examined. The basic biological effects of ionizing radiation, applications of ionizing radiation, and the established maximum permissible dose levels for the working environment have been presented.